

and environmental monitoring. Hence, scores of global companies are now experimenting with drone technology to provide state-of-the-art services, and various steps are being taken to make it more popular in the coming years. In the past few years, technologies such as machine learning, artificial intelligence (AI), GPS, and other new technologies have massively improved the functioning and efficiency of drones in the logistics industry. E-commerce and logistics companies are constantly trying to introduce more innovations among drones.

Scope for drone manufacturing industry in India

Currently, with the outbreak of Covid-19 pandemic, homeland security department has understood the requirement of blending security framework with this cutting-edge technology. The department's interest in drones has escalated massively for an impeccable security-proofing and surveillance.

In the past, there was a massive lack of regulation that posed a major impediment. It is now being slowly tackled with the introduction of civil aviation requirements. Moreover, the difficulty in obtaining drones for public sector firms will now decrease with the introduction of drones as a separate category in Government e-Marketplace (GeM).

Several drones are being used during Covid-19 pandemic, which will push its growth rate all over the world, especially in India. According to market experts, government organisa-

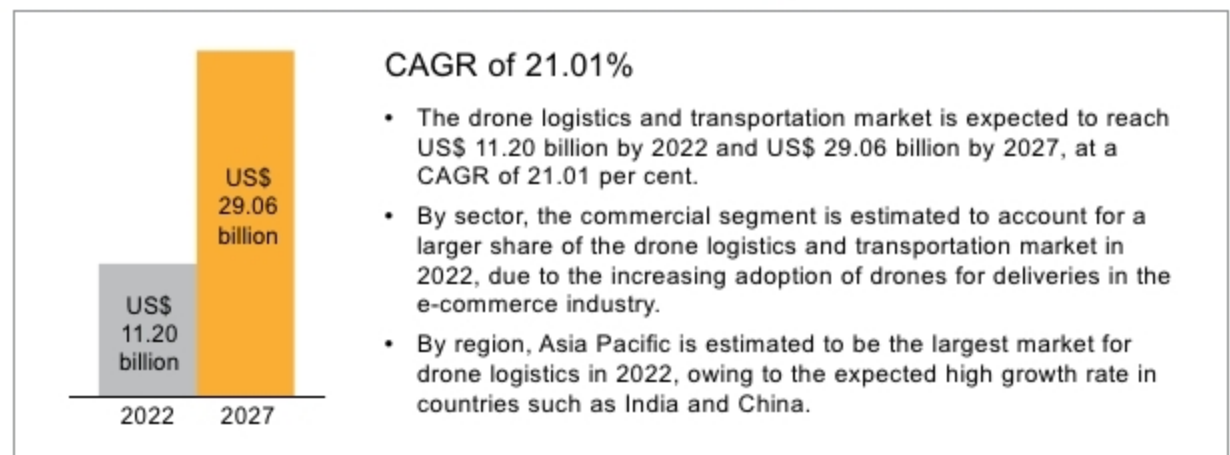


Fig. 1: Opportunities in drone logistics and transportation market (Credit: www.marketsandmarkets.com)

tions such as police force and defence would start buying drones in huge numbers. Therefore the DGCA drones in the surveillance sectors could witness rapid growth soon.

Amit Thokal, marketing manager at IdeaForge, states, "We are creating high-performance, safe, and autonomous unmanned aerial systems that enhance productivity and safety. Our aim is to increase the adoption of drones in India for public safety, security and surveillance, and emergency response and create an ecosystem that enhances the quality of life. We are India's one of the largest manufacturers of drones for defence, homeland security, and industrial applications. IdeaForge has over 90 per cent share in India in the security and surveillance market of homegrown drones. We are also a major player in surveying and mapping, precision agriculture, asset inspection, and disaster management."

Glenford V D'Souza, senior general manager - LYNX: Lawrence & Mayo, says, "With the prevailing Covid-19 situation, we are certain that the use of this versatile technology would be

better seen and experienced by the health authorities and the police. Our drones are extensively helping the law authorities in several parts of the city for providing surveillance and security in areas that were hard to reach."

Potential market areas of drone

The benefits of using a drone are varied and large. It reduces manpower and helps speed up operations, especially in the defence sector. Also, drones assist police in monitoring large crowds, tracking vehicles, and ensuring social distancing during Covid-19. Drones fitted with radioactive cameras/radiometric thermal imaging sensors can even be used to measure the temperature of individuals by flying at lower heights.

Drones, which are capable of flying distances ranging from two to four kilometres, are able to capture still images and videos to gather information about specific targets such as individuals, groups or environments. Their surveillance abilities enable surreptitiously gathering information about a target as captured from a distance or altitude.

The drone logistics and transportation market is segmented by solution—warehousing, infrastructure, shipping, or software; sector—military, or commercial; drone type—passenger drones, freight drones, or ambulance drones; and geography. Over time drones will become part and parcel of households and industries. Additionally, there are several other applications that are currently being explored to make these even more dynamic. For now, drones are being used mostly in the following sectors:

- Aerial photography for journalism and film making



Fig. 2: Drone logistics and transportation market growth rate by region during 2020-2025 (Credit: Mordor Intelligence)